POLYP DETECTION MEDICAL REPORT

Analysis Date: May 18, 2026

VIDEO INFORMATION

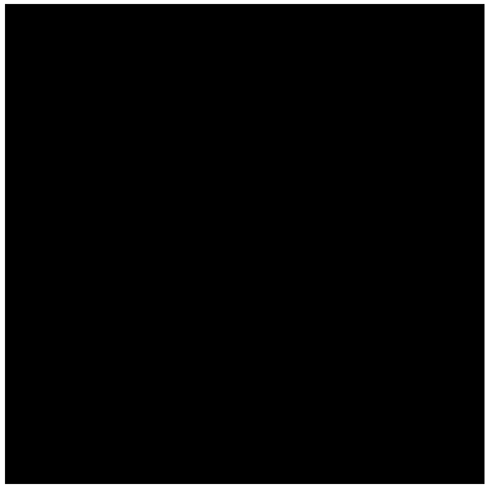
Video ID:	synthetic
Total Frames:	8
Frame Rate:	25.0 fps (assumed)
Duration:	0.3 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	0	✓
RT-DETR Detections	0	✓
MedSAM Detections	0	✓
Consensus (3-Model Agreement)	0	✓
Risk Classification	**Count**	**Percentage**
HIGH RISK	1	50.0%
LOW RISK	1	50.0%
TOTAL ANALYZED	2	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 - LOW

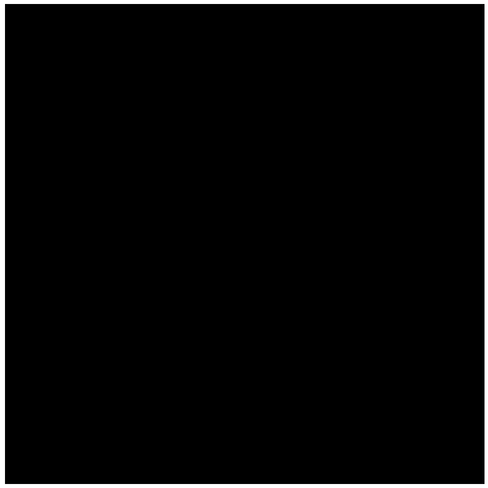


Feature	Value	Interpretation
Frame Index	0	Frame 0 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	LOW	AI Decision Tree result
Confidence Score	62.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #2 - LOW

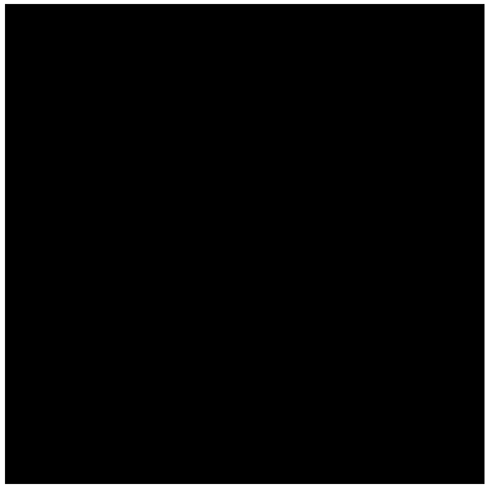


Feature	Value	Interpretation
Frame Index	1	Frame 1 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	LOW	AI Decision Tree result
Confidence Score	81.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #3 - LOW

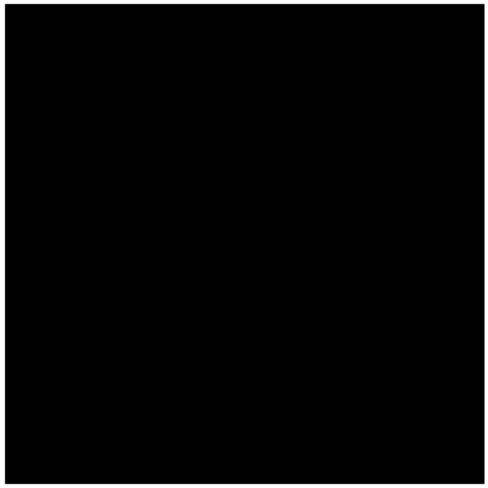


Feature	Value	Interpretation
Frame Index	2	Frame 2 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	LOW	AI Decision Tree result
Confidence Score	74.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #4 - LOW

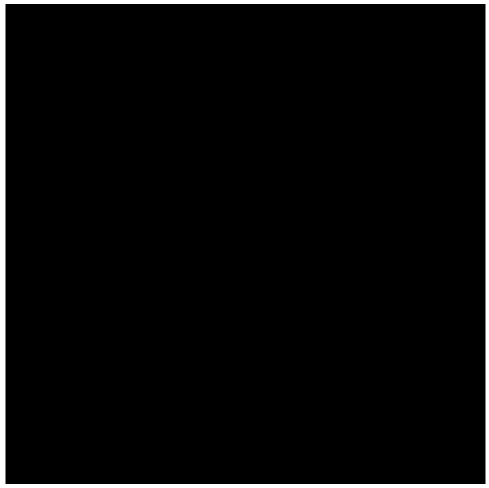


Feature	Value	Interpretation
Frame Index	3	Frame 3 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	LOW	AI Decision Tree result
Confidence Score	59.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

POLYP #5 - HIGH

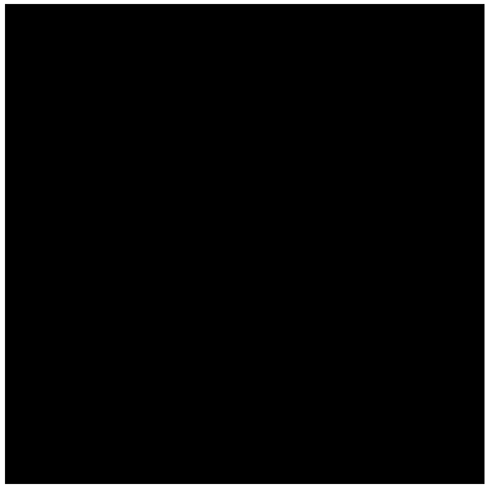


Feature	Value	Interpretation
Frame Index	4	Frame 4 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	HIGH	AI Decision Tree result
Confidence Score	91.0%	Model certainty level

Type Assessment: UNKNOWN lesion

High-confidence HIGH RISK detection - Requires close monitoring and potential intervention

POLYP #6 - HIGH

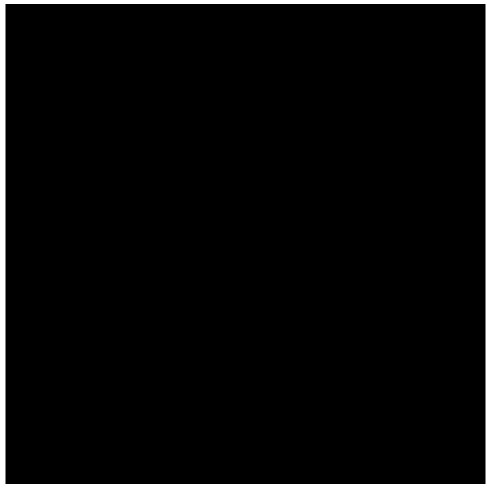


Feature	Value	Interpretation
Frame Index	5	Frame 5 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	HIGH	AI Decision Tree result
Confidence Score	88.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence HIGH RISK detection - Further evaluation recommended

POLYP #7 - LOW



Feature	Value	Interpretation
Frame Index	6	Frame 6 of 8
Detection Method	Multi-Model Consensus	YOLO + RT-DETR + MedSAM agreement
POLYP TYPE	UNKNOWN	Type confidence: 50.0%
Redness Score	0.000	Color-based vascularization
Radius (pixels)	0.0	Polyp size measurement
Texture Score	0.000	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Medical Risk Score	0.000	Combined heuristic assessment
Expert Classification	LOW	AI Decision Tree result
Confidence Score	35.0%	Model certainty level

Type Assessment: UNKNOWN lesion

Moderate-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:
A total of 7 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:
• HIGH RISK polyps: 0 (0.0%)
• LOW RISK polyps: 0 (0.0%)

Recommendation:
All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.
Technical Details:
• Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM)
• Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
• Classification: Cluster-specific Decision Tree experts
• Confidence: Based on deep learning + medical heuristics

AI-Based Polyp Detection System
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